

ERZURUM, Turkey

WMO: 170960

Lat: 39.953N

Lon: 41.190E

Elev: 5768

StdP: 11.88

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-13.1	-23.5	1.9	-18.7	-17.3	2.6	-13.0	24.3	22.0	22.1	20.8	2.0	220	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	31.7	87.8	58.2	84.7	57.9	82.3	57.9	62.5	79.9	61.0	78.3	59.5	76.7	10.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.7	82.1	69.8	54.0	77.0	68.3	52.8	73.6	67.2	31.3	80.2	30.0	77.8	28.9	76.4	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	20.0	17.9	DB	-24.2	91.9	6.8	3.3	-29.1	94.3	-33.0	96.2	-36.9	98.1	-41.8	100.5	(4)
(5)			WB	-24.4	65.8	6.7	3.0	-29.3	68.0	-33.2	69.7	-36.9	71.4	-41.8	73.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	14.1	16.7	29.1	42.1	50.9	58.3	66.1	66.9	57.4	46.9	33.3	19.7
	DBStd	19.93	12.15	11.97	10.25	6.46	5.03	4.64	4.70	4.36	5.34	5.92	8.22	12.80
	HDD50	4564	1112	931	648	245	50	2	0	0	8	128	501	939
	HDD65	8593	1577	1351	1113	687	437	206	42	30	234	561	951	1404
	CDD50	1620	0	0	0	9	78	250	498	524	229	32	0	0
	CDD65	174	0	0	0	0	0	5	75	89	5	0	0	0
	CDH74	3636	0	0	0	0	7	248	1351	1651	375	4	0	0
	CDH80	1076	0	0	0	0	0	25	423	564	64	0	0	0
Wind	WSAvg	6.7	4.9	5.4	7.2	8.0	7.4	7.2	8.3	8.3	6.9	6.2	5.7	4.9
Precipitation	PrecAvg	15.8	0.8	0.9	1.3	2.1	2.7	1.8	1.0	0.7	0.8	1.7	1.3	0.9
	PrecMax	25.1	2.2	3.5	3.2	5.8	6.1	4.4	3.7	2.5	3.0	4.1	3.2	2.1
	PrecMin	9.9	0.1	0.0	0.0	0.4	0.6	0.1	0.0	0.0	0.0	0.1	0.0	0.2
	PrecStd	3.1	0.5	0.6	0.7	1.1	1.2	1.0	0.7	0.6	0.7	1.0	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	42.7	57.3	68.3	74.9	82.8	91.3	91.5	85.7	73.6	57.6	48.3
		MCWB	34.1	36.4	42.7	49.1	54.4	58.5	59.4	58.4	54.7	51.0	43.1	38.8
	2%	DB	36.3	38.6	51.7	63.7	71.3	79.2	87.7	88.2	81.5	69.8	53.9	41.6
		MCWB	32.5	34.2	40.3	47.6	52.8	56.8	59.1	57.7	54.7	49.8	41.7	35.9
	5%	DB	34.1	35.8	47.6	59.3	68.1	76.9	84.5	86.0	78.5	66.0	50.3	38.9
		MCWB	31.1	32.3	39.0	45.5	51.0	56.1	58.8	57.6	53.5	48.8	40.3	34.6
	10%	DB	31.9	33.5	43.1	55.4	64.6	73.7	82.0	83.0	75.2	61.9	46.7	35.9
		MCWB	29.8	30.7	37.0	43.8	50.3	55.4	58.8	57.8	52.7	47.8	39.1	32.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.9	37.1	44.5	51.6	58.2	62.7	65.7	64.3	59.2	54.1	45.6	39.8
		MCDB	37.3	41.0	55.7	63.4	69.9	76.6	82.5	82.5	78.0	67.5	53.6	45.6
	2%	WB	33.0	34.4	41.5	48.6	54.9	59.8	62.9	62.0	57.2	51.8	43.4	36.7
		MCDB	35.4	37.5	49.5	60.6	66.5	74.8	80.3	80.3	74.9	65.5	51.0	40.1
	5%	WB	31.5	32.4	39.2	46.6	53.0	58.0	61.5	60.5	55.4	49.9	41.4	34.8
		MCDB	33.9	35.0	46.0	57.0	64.3	72.5	78.7	78.8	73.4	61.9	48.1	37.9
	10%	WB	29.5	30.6	37.1	44.7	51.3	56.5	60.1	59.2	53.9	48.4	39.7	32.8
		MCDB	31.7	33.0	42.5	53.9	62.2	70.6	77.0	77.8	71.5	59.7	45.9	35.3
Mean Daily Temperature Range	5% DB	MDBR	19.9	21.1	19.3	21.6	24.2	28.6	30.2	31.7	32.4	26.6	23.1	19.3
		MCDBR	14.8	16.2	23.3	29.5	30.7	33.9	34.4	35.9	38.4	35.2	30.2	20.3
		MCWBR	11.9	12.4	13.7	16.2	15.6	15.4	13.6	13.9	17.2	18.2	18.9	14.7
	5% WB	MCDBR	13.5	14.9	20.0	26.3	26.7	29.9	30.1	31.7	33.3	29.4	25.2	17.0
		MCWBR	11.3	11.6	12.3	15.3	14.8	15.3	13.6	14.1	16.0	16.1	16.5	12.5
Clear-Sky Solar Irradiance	taub		0.274	0.312	0.367	0.424	0.392	0.356	0.369	0.373	0.322	0.350	0.298	0.270
	taud		2.233	2.103	2.048	2.011	2.201	2.354	2.336	2.303	2.430	2.286	2.391	2.328
	Ebn at Noon		281	283	278	269	281	291	286	282	292	268	272	275
	Edh at Noon		30	41	48	54	45	39	39	40	32	34	26	26
All-Sky Solar Radiation	RadAvg		784	1145	1460	1546	1803	2238	2286	2045	1684	1096	748	624
	RadStd		83	96	144	160	136	128	122	88	102	89	102	99

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	+1.35	-1.03	N/A	N/A	N/A	+131	+49
	N/A	N/A	N/A	+1.15	-0.81	N/A	N/A	N/A	+160	+67

Nomenclature: See separate page